

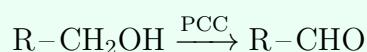
Chapter 8: Aldehydes, Ketones & Carboxylic Acids

Named Reactions Sheet + Tips & Tricks

Formula Sheet

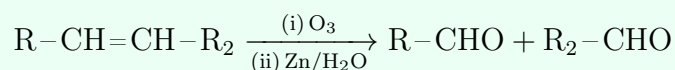
I. Preparation of Aldehydes and Ketones

From Alcohols (Oxidation)



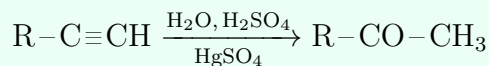
1° alcohol gives aldehyde (PCC stops at aldehyde stage); 2° alcohol with acidic K₂Cr₂O₇ gives ketone.

From Alkenes – Ozonolysis



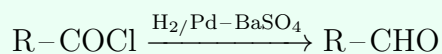
Reductive ozonolysis of an alkene cleaves the double bond into two carbonyl compounds.

From Alkynes – Hydration



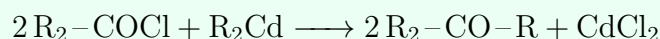
Follows Markovnikov's rule; terminal alkynes give methyl ketones (except acetylene itself, which gives acetaldehyde).

From Acid Chlorides – Rosenmund Reduction



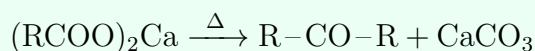
Poisoned (partially deactivated) Pd catalyst stops reduction at the aldehyde stage.

From Acid Chlorides – Dialkyl Cadmium



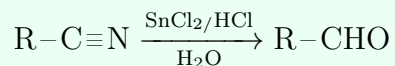
Reaction with dialkyl cadmium gives a ketone; the reagent is unreactive toward other functional groups, so it works even on halo-acid chlorides.

From Calcium Salts of Carboxylic Acids



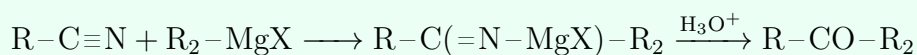
Dry distillation of calcium salt gives a symmetrical ketone.

From Nitriles – Stephen Reaction



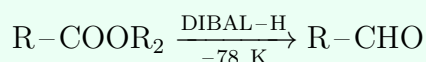
Reduction of nitrile with stannous chloride and HCl, followed by hydrolysis of the imine intermediate, gives an aldehyde.

From Nitriles – Grignard Reagent



Grignard addition to nitrile, followed by hydrolysis of the imine salt, gives a ketone.

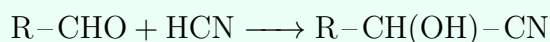
From Esters – Partial Reduction



DIBAL-H at low temperature stops ester reduction at the aldehyde stage (excess/room temperature gives the alcohol).

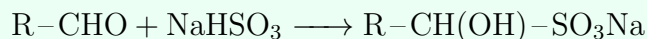
II. Chemical Properties – Nucleophilic Addition

Addition of HCN



Forms a cyanohydrin; base-catalysed (traces of KOH/CN[−] needed to generate nucleophile).

Addition of NaHSO₃

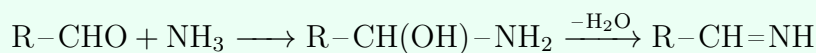


Forms a bisulphite addition compound (white crystalline solid); used to purify aldehydes/methyl ketones.

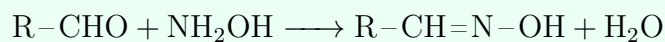
Addition of Grignard Reagent



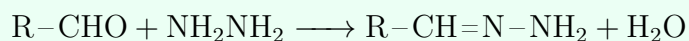
Aldehyde + Grignard gives a secondary alcohol; ketone + Grignard gives a tertiary alcohol.

Reaction with Ammonia

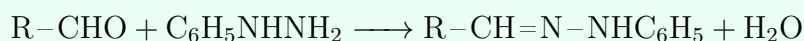
Forms an imine; ammonia derivatives (below) react similarly via the same addition-elimination pathway.

Reaction with Hydroxylamine

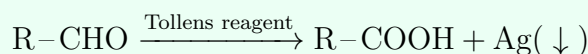
Forms an oxime.

Reaction with Hydrazine

Forms a hydrazone.

Reaction with Phenylhydrazine

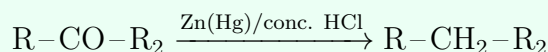
Forms a phenylhydrazone (used to characterise carbonyl compounds; with excess reagent on sugars, gives an osazone).

III. Oxidation, Reduction and Alpha-Hydrogen Reactions**Oxidation of Aldehydes**

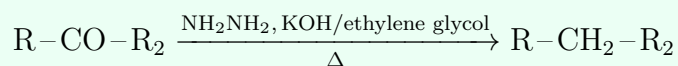
Tollens' reagent (ammoniacal AgNO_3) gives a silver mirror; Fehling's solution gives a brick-red Cu_2O precipitate – both distinguish aldehydes from ketones since ketones do not reduce these reagents.

Iodoform Reaction

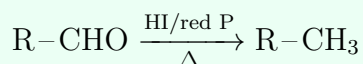
Positive for methyl ketones and acetaldehyde (compounds with a $\text{CH}_3\text{CO-}$ group), and for ethanol/secondary alcohols with a $\text{CH}_3\text{CH(OH)-}$ group (oxidised in situ first).

Clemmensen Reduction

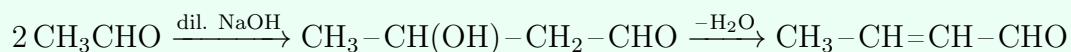
Reduces the carbonyl group to CH_2 under acidic conditions; unsuitable for acid-sensitive substrates.

Wolff-Kishner Reduction

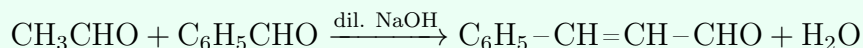
Reduces the carbonyl group to CH_2 under basic conditions – complementary to Clemmensen for base-sensitive substrates.

Reduction with Red P and HI

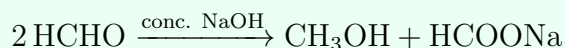
Complete reduction of the carbonyl to a methyl/methylene group.

Aldol Condensation

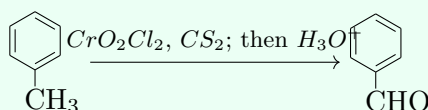
Requires alpha-hydrogen; base-catalysed addition gives the aldol, which dehydrates on heating to the alpha-beta unsaturated carbonyl compound.

Cross Aldol Condensation

Between two different carbonyl compounds, at least one having alpha-hydrogen; benzaldehyde (no alpha-H) here acts as the electrophile.

Cannizzaro's Reaction

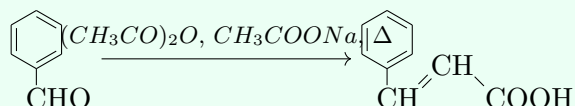
Self-oxidation-reduction of an aldehyde with *no* alpha-hydrogen; one molecule is oxidised to the carboxylate salt, the other reduced to the alcohol.

IV. Benzaldehyde – Special Reactions**Preparation from Toluene – Etard/Chromyl Chloride**

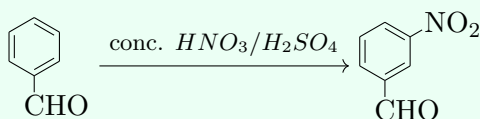
Lab preparation via oxidation with chromyl chloride (Etard reaction), through a chromium complex intermediate.

Benzoin Condensation

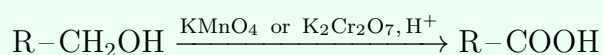
Cyanide-ion catalysed self-condensation of an aromatic aldehyde (two molecules, no α -H) gives an α -hydroxy ketone (benzoin).

Perkin's Reaction

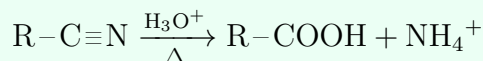
Aromatic aldehyde condenses with an acid anhydride (having α -H) in presence of the sodium salt of the acid, giving a cinnamic acid derivative.

Electrophilic Substitution – Nitration

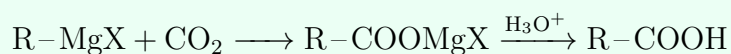
-CHO is a deactivating, meta-directing group.

V. Preparation of Carboxylic Acids**From Alcohols/Aldehydes – Oxidation**

Full oxidation of a 1° alcohol or aldehyde.

From Nitriles – Hydrolysis

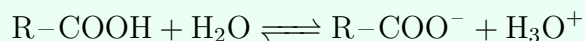
Acidic (or alkaline) hydrolysis of a nitrile gives the carboxylic acid (one carbon more than the parent alkyl halide).

From Grignard Reagent

Grignard reagent reacts with dry ice (solid CO_2), then acidic workup gives the carboxylic acid.

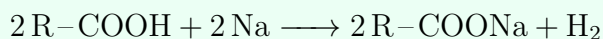
VI. Chemical Properties of Carboxylic Acids

Acidic Character



Carboxylic acids are more acidic than phenols and alcohols due to resonance stabilisation of the carboxylate ion.

Reaction with Active Metals

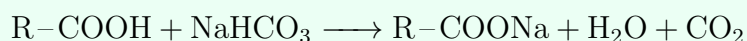


Brisk effervescence of H_2 gas.

Reaction with Alkalies

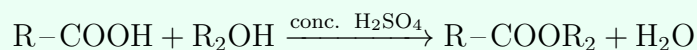


Reaction with Carbonates/Bicarbonates



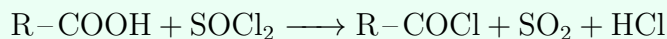
Brisk effervescence of CO_2 – classic test to distinguish carboxylic acids from phenols (phenols do not react with NaHCO_3).

Esterification



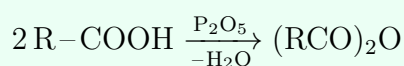
Fischer esterification; reversible.

Formation of Acid Chloride

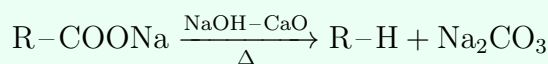


$\text{PCl}_3/\text{PCl}_5$ can also be used.

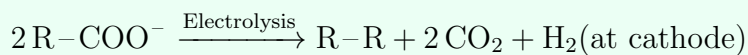
Formation of Acid Anhydride



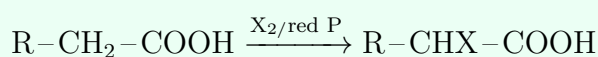
Dehydration of two carboxylic acid molecules.

Decarboxylation – Chemical (Soda Lime)

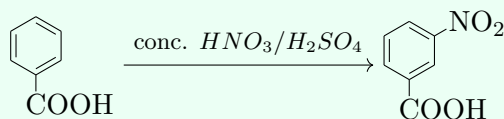
Sodium salt heated with soda lime removes CO_2 , giving an alkane one carbon shorter.

Kolbe's Electrolytic Decarboxylation

At the anode, carboxylate ions lose electrons and CO_2 , and the resulting radicals combine to form a symmetrical alkane.

Hell-Volhard-Zelinsky (HVZ) Reaction

Alpha-halogenation of a carboxylic acid having alpha-hydrogen, using Cl_2/Br_2 with red phosphorus.

Substitution on Benzoic Acid Ring – Nitration

$-\text{COOH}$ is a deactivating, meta-directing group.

Tips, Tricks & Hacks

- **Distinguishing aldehydes from ketones:** Tollens' test (silver mirror), Fehling's test (brick-red ppt.), and Schiff's test all work only on aldehydes – ketones give a negative result on all three.
- **Iodoform-positive compounds:** any $\text{CH}_3\text{CO-R}$ (methyl ketone) or CH_3CHO , plus ethanol and any $\text{CH}_3\text{CH(OH)-R}$ alcohol (oxidised to the methyl ketone/aldehyde *in situ* by the I_2/NaOH before the haloform step).
- **Cannizzaro vs. Aldol:** no alpha-hydrogen (e.g., HCHO , $\text{C}_6\text{H}_5\text{CHO}$, $(\text{CH}_3)_3\text{CCHO}$) → Cannizzaro (self-redox) with strong base; alpha-hydrogen present → Aldol condensation with dilute base.
- **Clemmensen vs. Wolff-Kishner:** pick Clemmensen ($\text{Zn-Hg}/\text{HCl}$, acidic) for base-sensitive substrates, Wolff-Kishner ($\text{NH}_2\text{NH}_2/\text{KOH}$, basic) for acid-sensitive substrates – both fully reduce C=O to CH_2 .
- **Acid strength order:** carboxylic acid > carbonic acid > phenol > water > alcohol – this is why carboxylic acids liberate CO_2 from NaHCO_3 but phenols do not.
- **HVZ reaction** needs an alpha-hydrogen on the carboxylic acid, exactly like Aldol condensation on aldehydes/ketones – both are enolisation-driven.

- Rosenmund reduction uses a *poisoned* Pd catalyst (Pd-BaSO_4 , sometimes with a trace of sulfur/quinoline) specifically to prevent over-reduction of the aldehyde to the alcohol.